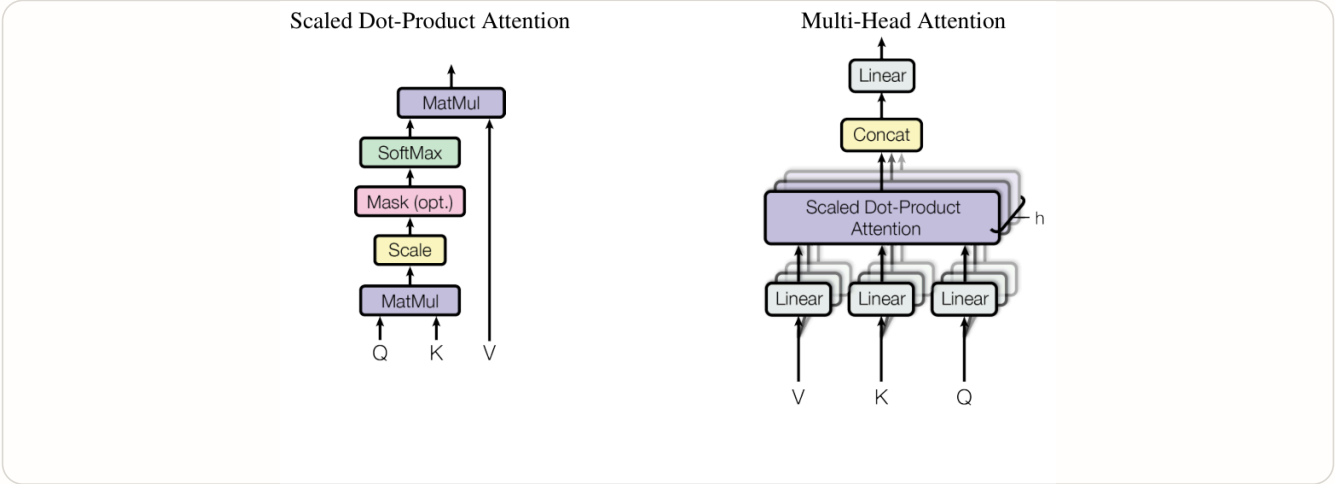


Attention Is All You Need

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(left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Task

Modeling Dependencies Without Recurrence

- Attention models dependencies regardless of distance in input or output sequences.
- The Transformer relies entirely on attention, eschewing recurrence to draw global dependencies.
- Future plans include extending the model to images, audio, and video modalities.

Challenges

Sequential Bottlenecks in Previous Models

- Recurrent models preclude parallelization within training examples.
- Previous convolutional models make it difficult to learn dependencies between distant positions.
- A single convolutional layer does not connect all pairs of input and output positions.

Proposed method

Pure Self-Attention Architecture Design

- The Transformer uses stacked self-attention and point-wise fully connected layers for both encoder and decoder.
- It employs $h=8$ parallel attention heads, each with dimension $dk=dv=64$, to attend to different subspaces simultaneously.
- Self-attention connects all positions with constant sequential operations, unlike recurrent layers requiring $O(n)$ operations.

Main results

State-of-the-Art Translation and Parsing Results

- The big Transformer model achieves a state-of-the-art BLEU score of 28.4 on WMT 2014 English-to-German translation.
- On WMT 2014 English-to-French, the model reaches a BLEU score of 41.0 at less than one-quarter the training cost.